

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First / Second Semester of B. Tech. Examination (All branch)

MAY 2014

EE 101 Fundamentals of Electrical & Electronics Engineering

Date: 22.05.2014, Thursday Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1 (a) State and explain Kirchhoff's laws with necessary diagram. [4]

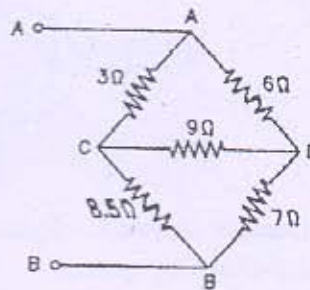
(b) Define: [3]

- (i) Electric field intensity
- (ii) Electric potential
- (iii) Electric flux density

Q – 2 (a) Show that if α_{t1} is the temperature coefficient of resistance of a conductor at $t_1^\circ\text{C}$, the temperature coefficient of resistance at $t_2^\circ\text{C}$ is given by [7]

$$\alpha_{t2} = \frac{1}{\frac{1}{\alpha_{t1}} + (t_2 - t_1)}$$

(b) Using delta/star transformation, find the value of the equivalent resistance between points A and B in given fig. [7]



OR

Q – 2 (a) Explain magnetic hysteresis with the help of B-H curve. [7]

(b) Give the Comparison of Electric circuit and magnetic circuit. [7]

Q – 3 (a) An alternating current having an equation $I = 141.4\sin 314t$. [7]

Find (i) Frequency (ii) RMS value (iii) Average value (iv) Instantaneous value of current When 't' = 2ms.

(b) An alternating voltage of $V = 100\sin 376.8t$ is applied to a coil having resistance $6\ \Omega$ and inductance of 21.22 mH. [7]

Determine (i) current through the coil

(ii) power dissipated in the coil

OR

- Q-3 (a) Give comparison of series and parallel resonance in R-L-C A.C circuit. Derive expression of quality factor for series R-L-C circuit. [7]
- (b) Derive the relationship between α and β . [7]

SECTION - II

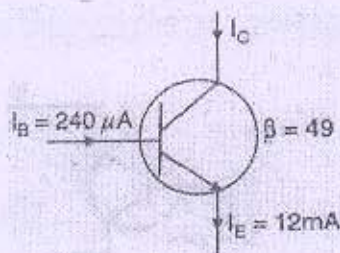
- Q-4 Define and explain the following : [7]

(i) frequency (ii) form factor (iii) peak factor (iv) time period (v) amplitude (vi) instantaneous value (vii) cycle

- Q-5 (a) (i) Convert $(846)_{10}$ number into Hexa-decimal number. [7]
- (ii) Convert $(101101101111011)_2$ number into Octal number.
- (iii) perform $(4562)_8 + (2317)_8$
- (b) Define Root mean square (RMS) value. Express RMS value in case of sinusoidal current and half wave rectifier. [7]

OR

- Q-5 (a) Explain the Zener diode as voltage stabilizer. [7]
- (b) Find the α rating of the transistor shown in below figure. Hence determine the value of I_c using both α and β rating of the transistor. [7]



- Q-6 (a) Give the basic difference between digital and analog system. Also state the advantages of digital over analog system. [7]
- (b) Define capacitance. Derive an expression for the capacitance of a parallel plate capacitor. [7]

OR

- Q-6 (a) Compare series circuits with parallel circuits. [7]
- (b) What do you understand by the term "resistance"? State and explain factors affecting resistance of a conductor. [7]
